

Problem-Solution Fit

Date	2 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] • Smallholder farmers seeking optimal yields. • Agricultural researchers & soil analysts. • Government policy planners.	6. CUSTOMER LIMITATIONS [CL] • Limited budget for commercial laboratory tests. • Reliance on entry-level smartphones or local kiosks. • Intermittent internet connection.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] • Government labs: High accuracy but slow turnaround (weeks). • Crop dealers: Biased towards selling specific fertilizers. • Simple calculators: Ignore climate dynamics.
2. PROBLEMS / PAINS + FREQUENCY [PR] • Low yields / crop failures due to improper matching (Seasonal). • Soil degradation from over-fertilization (Constant). • Manual data interpretation is slow.	9. PROBLEM ROOT / CAUSE [RC] • Rapid changes in climate envelopes. • Lack of affordable, real-time testing software. • Farmers lack direct access to machine learning insights.	7. BEHAVIOR + INTENSITY [BE] • Relies on historical crops (High Intensity). • High pesticide/fertilizer usage as a generic safety net.
3. TRIGGERS TO ACT [TR] • Soil testing yields confusing numbers. • Drastic changes in seasonal rainfall levels. • Financial pressure to improve farm outputs.	10. YOUR SOLUTION [SL] OptiCrop Smart Agricultural Production Engine: • ML-based crop recommendation. • Real-time soil compatibility checks. • Visual dashboards.	8. CHANNELS OF BEHAVIOR [CH] ONLINE: • Accessing agricultural advisory web portals. • Looking up crop yields online.
4. EMOTIONS BEFORE / AFTER [EM] • Before: Anxious about harvest, frustrated by complex raw numbers. • After: Confident, relieved, empowered with advice.		OFFLINE: • Physical farmer markets. • Local agricultural cooperative meetups.